Effects of Neutron Irradiation on Nd–Fe–B Magnetic Properties

J. R. COST, R. D. BROWN, A. L. GIORGI, AND J. T. STANLEY

***Abstract*—Nd–Fe–B permanent magnets from two different manufacturers were irradiated in the Omega West reactor at Los Alamos National Laboratory, Los Alamos, NM, with fast neutrons at temperatures of 426 K (153°C) and 350 K (77°C) to fluences of 5.0×10^{16} N/cm^2 and 6.1×10^{16} N/cm^2, respectively. At intervals during the irradiation the samples were removed from the reactor and the remanence measured at room temperature. The initial loss of remanence for irradiation at 426 K was ten percent for a fluence of 10^{15} N/cm^2. At 350 K the initial loss rate was roughly half this value. The loss rates were nearly the same for samples from the two different manufacturers. These losses are due to the irradiation since the remanence does not decay with annealing at 426 K. Remagnetization after irradiation results in full recovery of the remanence and roughly a 20-percent increase in the coercivity. Evidence from this experiment suggests that the primary mechanism for loss of remanence is nucleation of reverse domains by the collision cascade and subsequent growth into the original domain.**

I. Introduction

The HIGH remanence, large energy product, and relatively low price of the recently developed Nd–Fe–B permanent magnetic materials has suggested the possibility of their use in a variety of applications, including particle accelerators. The original use of permanent magnets in particle accelerators followed the development of the high-coercivity Sm–Co magnets in the mid 1960's.

Use of permanent magnets in locations where they are exposed to a flux of high-energy particles can lead to a degradation of their magnetic properties. There have been relatively few studies on the effects of irradiation on the properties of permanent magnets. Extensive studies in 1959 [1] and 1961 [2] showed effects of neutron irradiation on the best magnets available at that time. Use of the Sm–Co magnets in high-radiation environments required investigation of radiation effects upon their magnetic properties. The first of these studies [3], conducted at the Los Alamos Meson Physics Facility, showed a two percent or less loss of remanence for $SmCo_5$ magnets irradiated to 1.1×10^{18} N/cm^2.

Further investigations on the Sm–Co alloy magnets were reported in successive years by CERN [4], Jülich [5], and TRIUMF [6]. Only the latter of these studies reported radiation effects on the newer Nd–Fe–B magnets. This investigation showed that a Nd–Fe–B magnet irradiated with 500 MeV protons sustained a loss of remanence of 55 percent after a dose of roughly 10^{14} p/cm^2 at 343 K (70°C). Such a large loss after a relatively short exposure would prevent most applications of these materials in radiation environments. In the present study we have irradiated Fe–Nd–B magnets with reactor neutrons to learn more about this problem.

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II. Experimental

A. Sample Preparation and Magnetic Measurements

Materials were received from two different manufacturers, Crucible Materials Corporation and I. G. Technologies. The materials were received in the form of blocks approximately 25 mm on a side. Small samples were cut from these blocks using a diamond impregnated wheel on a low-speed cutting machine. The samples were 7 mm long with a square cross section 2×2 mm.

After cutting the samples the complete magnetization versus field hysteresis loop was measured using a vibrating sample magnetometer and a superconducting magnet. All the samples were measured at 250 K, the upper operating temperature of this magnetometer. Curves of magnetization versus field at 250 K for samples of the two different materials were measured before and after the irradiation sequence to be described. For the post-irradiation measurements the second quadrant of the hysteresis loop was measured initially. All hysteresis loop measurements were terminated in the first quadrant after applying fields of at least 55 kOe. Saturation, both for pre- and post-irradiation samples and for the magnets of both manufacturers, occurred at roughly 40 kOe.

At selected times during the irradiation the samples were removed from the reactor and the open-circuit remanence was measured at room temperature. Measurements were made by moving the sample through a 500-turn coil. The voltage produced as the magnet moved through the pickup coil was sampled by a Keithly Model 194 high-speed digital voltometer at a rate of 1 kHz. These voltage readings were sent to a computer which numerically integrated them to calculate the area under the voltage–time curve and thus a quantity proportional to the open-circuit remanence. Each value of the remanence which is reported is an average of five or more separate

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measurements. The standard deviation of these measurements is typically less than two percent of the remanence.

B. Irradiation Procedure

The samples were irradiated in the Omega West reactor in the epithermal cadmium port using a rabbit facility which allows samples to be removed during reactor operation. The neutron spectrum at this port has recently been determined by the unfolding of activation foil data [7]. Based upon this spectrum the fast neutron flux is estimated to be 4×10^{12} N/cm^2 s. Note that this flux is lower by a factor of five than values previously reported [8]. Thermal neutrons are important in irradiation of Nd–Fe–B because the (n, α) reaction which occurs with ^{10}B can produce extensive damage not accounted for by the fast neutrons alone. In this study, cadmium shielding around the port attenuated the thermal neutron flux so that the energy spectrum was typical for reactor neutrons but with energies less than 5 eV greatly attenuated.

The samples were placed in a polyethylene capsule which was fixed to the end of a chromel–alumel thermocouple and lowered into the rabbit tube using the thermocouple. The hot junction of the thermocouple was attached directly to a dummy sample of the same material and size as the actual samples. At the end of the desired irradiation time the samples were removed from the rabbit tube and the remanence wsa measured. Subsequently, the samples were reinserted into the reactor for further irradiation. The activity of the samples was about 1 R/h at contact immediately after a 1-h irradiation, decreasing to about 100 mR/h in about 5 min. Measurements and sample manipulation were carried out behind shielding to minimize exposure to the operator.

Two different irradiation temperatures were achieved, depending on the atmosphere in the rabbit tube. With the rabbit tube filled with helium the sample temperature reached 350 K (77°C) $\pm$ 1 K, and with the tube filled with air the sample temperature was 426 K (153°C) $\pm$ 2 K for irradiation times of 30 min or more. The time required to reach steady-state temperature was about 15 min in the latter case and about 5 min in the former.

III. Results and Discussion

Fig. 1 shows the normalized loss of remanence with irradiation time for the two different materials irradiated at the two temperatures. The preirradiation remanence values given in Table I were measured using the vibrating sample magnetometer at 250 K. As may be seen in Fig. 1, significant losses of magnetization start immediately with irradiation. For irradiation at 350 K the initial loss was roughly five percent for a fluence of 10^{15} N/cm^2. At 426 K it was roughly twice this rate. Sm–Co alloy magnets do not show similar losses until fluences of greater than 10^{18} N/cm^2 [3]–[6].

Note that for these particular samples from two different manufacturers there are only slight differences in the rates of loss of magnetization at a given temperature; however, there are large differences for the two different irradiation temperatures. To determine what portion, if any, of the remanence loss is due to annealing effects at the higher temperature, nonirradiated magnetized samples from each manufacturer were isothermally annealed at 423 K (150°C), and the remanence was measured during this anneal. It was found that the samples showed no long-term decay of remanence during this anneal for times up to 15 300 s. (A short-term decay of roughly 2 percent, which was complete after the first 200 s of the anneal, was observed. This effect is previously known and is classified as an irreversible loss which is recoverable upon remagnetization [9].)

It is desirable to analyze the decay kinetics to learn more about the mechanism(s) involved. Qualitatively, the kinetics at 426 K appear to be governed by at least one fast and one slow process, with the fast process saturating after roughly 80 percent of the decay has occurred. In agreement with the previous observation, attempts to actually fit the data to various single-process models were unsuccessful. From this we deduce that to predict both the fast and the slow decay, more than one mechanism must be involved.

It is interesting to consider why the decay occurs so much faster at higher temperature, roughly twice as fast initially and more than three times as fast when the remanence is 70 percent of its initial value. This may be

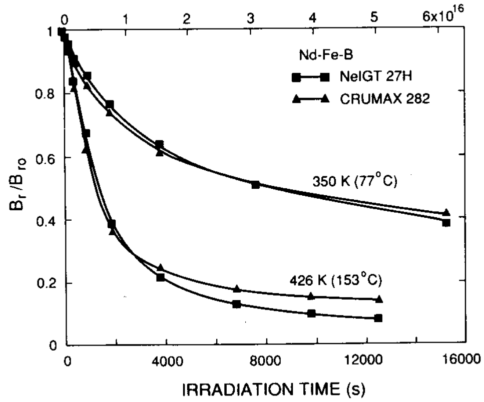

Fig. 1. Retained fraction of preirradiation remanence versus irradiation time and fluence for Nd–Fe–B samples of two different manufacturers irradiated with reactor neutrons at 350 and 426 K.

TABLE I
Preirradiation Remanence of Nd-Fe-B Samples at 250 K

Material	Remanence (kG) 350 K[a]	426 K[a]
NeIGT 27H	10.51	10.49
CRUMAX 282	9.94	10.43

[a]Denotes the temperature at which the sample was later irradiated.

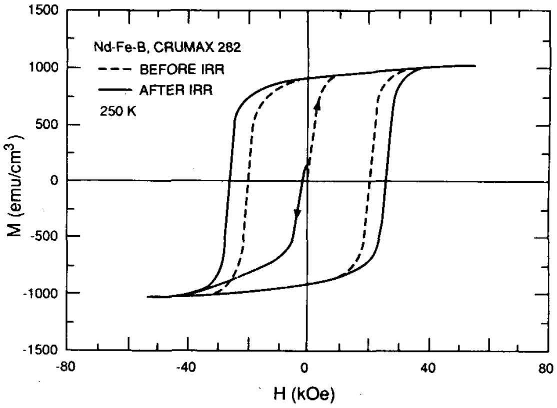

Fig. 2. Hysteresis loops at 250 K for CRUMAX 282 sample before and after irradiation to 6.1×10^{16} N/cm^2 at 426 K (153°C). Arrows show start of measurements.

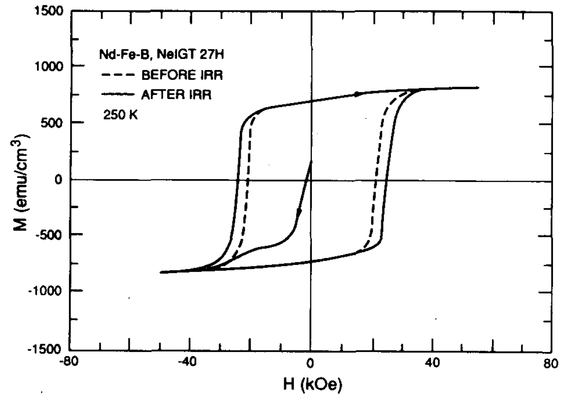

Fig. 3. Hysteresis loops at 250 K for NeIGT 27H sample before and after irradiation to 6.1×10^{16} N/cm^2 at 426 K (153°C). Preirradiation measurements were on magnetized sample and were started in first quadrant. Post-irradiation measurements started in second quadrant.

related to the relatively low Curie temperature of Nd–Fe–B, which is near 585 K (312°C). The decay process at 426 K may be more efficient than that at 350 K because a collision cascade has a higher probability of nucleating a reverse domain when the temperature is closer to the Curie temperature. This kind of behavior would be expected if classical nucleation theory controls the nucleation and the critical domain size decreases as temperature increases toward the Curie temperature. Independent of the details of the nucleation process, the evidence suggests that the fast and major part of the decay is associated with the mechanism of irradiation-induced nucleation and subsequent growth of reverse domains. The mechanism for the slow part is not known, but it may be associated with movement of domain walls to enlarge the reverse domains. Clearly, further investigation is required before mechanisms can be fully understood.

Figs. 2 and 3 show the hysteresis loops measured at 250 K for samples from the two different manufacturers before and after irradiation at 426 K (153°C) for the total irradiation time shown in Fig. 1. Notice that the remanence, which after irradiation has dropped to less than 20 percent of the preirradiation value for both of the samples, is restored to its original value by the remagnetization. This is good evidence that the radiation damage has not been extensive enough to change the metallurgical structure of these materials appreciably. Thus since the structure is unchanged, the same magnetic moment per unit volume is attainable when the samples are fully magnetized. It is interesting to note in Figs. 2 and 3 that the coercivity of the samples of both manufacturers has been increased roughly 20 percent by the irradiation. This increase is presumably due to the unannealed radiation damage. This damage in the form of collision cascdes may tend to either pin domain walls or to make it more difficult to nucleate new domains aligned in the direction of magnetization.

Taking the elastic scattering cross section to be about 3×10^{-24} cm^2 gives about 4×10^{-8} for the atomic fraction of primary knock-on events in 1 h. Estimating the number of atoms displaced by a primary knock-on event as about 300, then the average number of displacements per atom per hour is roughly 10^{-5}. This low fraction is in agreement with the previous discussion, indicating that the radiation dose is too low to modify the metallurgical structure. It is evidence that the decay is caused by a mechanism in which the spin of a large number of atoms is reversed, even though only a small number of atoms are displaced by the collision cascade. Again, a model for this is the radiation-induced nucleation of a reverse domain with subsequent growth throughout the original domain. This nucleation process must be quite inefficient since if we assume equiaxed domains of 1 μm, irradiation to a level of 10^{-5} average displacements per atom produces about 10^3 cascades per domain.

IV. Conclusion

1) The initial loss of remanence is nearly twice as fast for irradiation at 426 K (roughly 10 percent for a fluence of 10^{15} N/cm^2) than at 350 K.

2) This remanence loss is a result of the irradiation and is not due to annealing at 426 K during the irradiation.

3) Remagnetization after irradiation results in full recovery of the remanence and roughly a 20-percent increase in the coercivity.

References

[1] R. S. Sery, D. I. Gordon, and R. H. Lundsten, "Nuclear radiation effects in permanent magnets," US Naval Ordnance Laboratory, White Oak, Silver Spring, MD, NAVORD Rep. 6276, Apr. 1959.

[2] R. S. Sery, R. H. Lundsten, and D. I. Gordon, "Radiation damage thresholds for permanent magnets," US Naval Ordnance Laboratory, White Oak, Silver Spring, MD, NOLTR 61-45, June 1961.

[3] R. D. Brown, E. D. Bush Jr., and W. T. Hunter, "Radiation effects on samarium-cobalt permanent magnets," LA-9437-MS, July 1982.

[4] F. Coninckx, W. Naegele, M. Reinharz, H. Schoenbacher, and P. Seraphin, "Radiation effects on rare-earth cobalt permanent magnets," CERN/SPS, TIS-RP/IR/83-07, Feb. 1983.

[5] H. Spitzer and A. Weller, "Magnetisierungsverlust von Samarium-Kobalt Permanentmagneten in hohen Neutronenfeldern," SNQ 1 N/BH 22 05 84, May 1984.

[6] E. W. Blackmore, "Radiation effects of protons on samarium-cobalt permanent magnets," *IEEE Trans. Nucl. Sci.*, vol. NS-32, pp. 3669–3671, Oct. 1985.

[7] R. D. Brown, to be published.

[8] J. R. Cost, R. D. Brown, A. L. Giorgi, and J. T. Stanley, in *High Performance Ferromagnetic Materials* (*Materials Res. Soc. Symp. Proc.*), S. G. Sankar, Ed. Pittsburgh, PA: Materials Res. Soc., vol. 96, 1987, pp. 321–327.

[9] J. E. Gould, *Permanent Magnets and Magnetism*, D. Hadfield, Ed. New York: Wiley, 1962, ch. 10, pp. 443–471.

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